SQL PROJECT

EMPLOYEE INFORMATION SYSTEM

Project **Overview**

I'll be presenting my introductory SQL project, built around an Employee Management System.

This project is designed to demonstrate how basic SQL queries can be applied to manage and analyze employee data.

The goal of this project is simple: to practice and showcase fundamental SQL operations using a small, structured dataset.

I created two tables—Employees and Departments—and applied queries such as SELECT, WHERE, LIMIT, COUNT, SUM, MAX, JOINs, and even a basic subquery.

My
Choice of Dataset

I chose the Employee Management System dataset because it’s relatable and practical. Every organization has employees and departments, so this dataset makes it easy to understand how SQL can be applied in real-world scenarios. It also allowed me to cover all the basic query types in a meaningful way.

Show all employee names and their salaries

```sql
-- Show all employee names and their salaries.

SELECT Name
FROM employees;
```

Name
Alice Khan
Bilal Ahmed
Clara Das
David Roy
Emma Saha
Farhan Ali
Grace Paul
Henry Das

Find employees who work in the Finance department

```sql
-- Find employees who work in the Finance department.

SELECT Name
FROM employees
WHERE DepartmentID = 102;
```

Name
Bilal Ahmed
Farhan Ali

Display the first 3 employees from the Employees table

```sql
-- Display the first 3 employees from the Employees table.

SELECT Name
FROM employees
LIMIT 3;
```

Name
Alice Khan
Bilal Ahmed
Clara Das

Count how many employees are in the IT department

```sql
-- Count how many employees are in the IT department.

SELECT COUNT(*) AS Total_IT_Employees
FROM Employees
WHERE DepartmentID = 103;
```

Total_IT_Employees
2

Find the total salary paid to employees in the HR department

```sql
-- Find the total salary paid to employees in the HR department.

SELECT SUM(Salary) AS Total_Salary
FROM employees
WHERE DepartmentID = 101;
```

Total_Salary
105000

Find the highest salary among all employees

```sql
-- Find the highest salary among all employees.

SELECT MAX(Salary) AS Highest_Salary
FROM employees;
```

Highest_Salary
75000

Show employee names along with their department names

```sql
-- Show employee names along with their department names.

SELECT e.Name, d.DepartmentName
FROM Employees AS e
INNER JOIN Departments AS d
ON e.DepartmentID = d.DepartmentID;
```

Name	DepartmentName
Alice Khan	HR
Bilal Ahmed	Finance
Clara Das	HR
David Roy	IT
Emma Saha	Marketing
Farhan Ali	Finance
Grace Paul	Marketing
Henry Das	IT

Show all departments and their employees (including departments with no employees)

```sql
-- Show all departments and their employees (including departments with no employees).

SELECT d.DepartmentName, e.Name
FROM departments AS d
LEFT JOIN employees AS e
ON d.DepartmentID = e.DepartmentID;
```

DepartmentName	Name
HR	Clara Das
HR	Alice Khan
Finance	Farhan Ali
Finance	Bilal Ahmed
IT	Henry Das
IT	David Roy
Marketing	Grace Paul
Marketing	Emma Saha

Find employees who earn more than the average salary

```sql
-- Find employees who earn more than the average salary.

SELECT Name, Salary
FROM employees
WHERE Salary > (SELECT AVG(SALARY) AS Average_Salary
FROM employees);
```

Name	Salary
Bilal Ahmed	60000
David Roy	70000
Farhan Ali	62000
Henry Das	75000

Show the names of employees hired after 2019 who work in Marketing

```sql
-- Show the names of employees hired after 2019 who work in Marketing.

SELECT Name, DepartmentID
FROM employees
WHERE DepartmentID = 104 AND HireDate > '2019-12-31';
```

Name	DepartmentID
Grace Paul	104

Key
Learnings

Through this project, I learned how to structure tables, insert data correctly, and connect tables using joins. I also practiced writing queries that answer real business questions, like finding total salaries, highest earners, or employees hired after a certain date. These skills form the foundation for more advanced SQL work.

Challenges **Faced**

One of the main challenges I faced was handling date formats and ensuring queries ran without syntax errors. I also had to carefully design the dataset so that every query had a meaningful output. Overcoming these challenges gave me confidence in working with SQL more independently.

Conclusion

In conclusion, this project gave me hands-on experience with SQL basics in a structured, practical way. It showed me how powerful even simple queries can be in managing and analyzing data. Going forward, I plan to expand this project with more advanced queries and larger datasets. Thank you for listening.

Thank You!

for taking time to explore this project